

Chapter 1 – Introduction to Software Development Life Cycle (SDLC)

Learning Objectives

- Understand the purpose of SDLC.
- Explain each phase of the SDLC.
- Recognize the benefits of following a structured software development process.
- Identify the roles involved in an SDLC project.

What is SDLC?

Software Development Life Cycle (SDLC) is a structured process used to plan, analyze, design, develop, test, deploy, and maintain software systems. It provides a repeatable framework that helps teams deliver high-quality software on time and within budget.

Why SDLC is Important

- Improves software quality
- Reduces project risks
- Improves communication among stakeholders
- Enables predictable planning and budgeting
- Reduces maintenance cost
- Ensures customer satisfaction

SDLC Phases

- 1. Planning** – Define scope, objectives, stakeholders, budget, schedule, and risks.
- 2. Requirements Engineering** – Gather, analyze, validate, and document business and technical requirements.
- 3. System Design** – Prepare High-Level Design (HLD) and Low-Level Design (LLD), architecture, database design, and interfaces.
- 4. Development** – Implement the solution using coding standards, version control, code reviews, and unit testing.
- 5. Testing** – Verify and validate the software through functional, integration, system, performance, security, and user acceptance testing.
- 6. Deployment** – Release the application to production using deployment and rollback strategies.
- 7. Maintenance** – Monitor, fix defects, optimize performance, and enhance features after release.

Key Roles

Business Analyst, Product Owner, Project Manager, Solution Architect, Developers, QA Engineers, DevOps Engineers, Scrum Master, Operations Team, and End Users all contribute throughout the SDLC.

Advantages

- Better project planning
- Higher software quality
- Easier maintenance
- Improved traceability
- Better collaboration
- Reduced development cost over time

Real-World Example

A bank developing an online banking system follows SDLC by first defining business goals, gathering customer requirements, designing secure architecture, developing features, testing thoroughly, deploying using CI/CD, and maintaining the application through regular updates and monitoring.

Best Practices

- Involve stakeholders early
- Document requirements clearly
- Review designs before coding
- Automate testing and deployments
- Perform code reviews
- Continuously monitor production systems

Chapter Summary

SDLC provides a disciplined approach to software development that improves quality, minimizes risk, and ensures software meets business objectives. Every successful software project relies on a well-defined SDLC process.